

Computer Networks - Module 3

TCP/UDP and Transport Layer

The transport layer is responsible for end-to-end communication between hosts. The two main protocols are TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).

TCP characteristics:

- Connection-oriented (three-way handshake: SYN, SYN-ACK, ACK)
- Reliable delivery with acknowledgements and retransmission
- Flow control via sliding window
- Congestion control (slow start, congestion avoidance)

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UDP characteristics: connectionless, no guarantee of delivery or ordering, lower overhead, used for streaming and DNS.

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Port numbers

Well-known ports range from 0-1023 (e.g. HTTP=80, HTTPS=443, FTP=21, DNS=53). Registered ports range from 1024-49151. Dynamic/private ports range from 49152-65535. A socket is defined by the combination of IP address and port number.

Wireshark capture tip: filter `tcp.port==80` to isolate HTTP traffic, or `udp.port==53` for DNS queries during lab analysis.